Computational Social Science Term Paper

Framing and Sentiment Analysis of Headlines on US Intervention in Venezuela

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Abstract

This study applies a multilingual computational pipeline to examine how news media across four world regions - the United States, Europe, Latin America, and Venezuela — framed the news on Operation Absolute Resolve, carried out on 3 January 2026, when US military forces conducted a raid on the presidential residence in Caracas that resulted in the capture and transfer into US custody of Venezuelan president Nicolás Maduro. A total of 1,000 news headlines were collected via the MediaCloud API covering the period 2–10 January 2026 for the analysis. To capture different dimensions of media framing, multiple complementary methods are employed. Sentiment is assessed using two transformer-based multilingual models (XLM-roBERTa and NLP Town BERT base multilingual uncased). Emotional tone is further examined using the NRC Emotion Lexicon, enabling a more detailed analysis of affective content. In addition, framing is explored through bigram analysis, while Latent Dirichlet Allocation (LDA) is used to identify underlying thematic structures. Findings reveal significant cross-regional variation in both sentiment and emotional tone, consistent with expectations from framing and agenda-setting theories. US media adopts the most polarized framing, emphasizing regime change narratives. European media contextualizes Venezuela within Trump's broader foreign policy agenda. Latin American coverage foregrounds sovereignty and humanitarian concerns, while Venezuelan media exhibits a predominantly neutral, reactive, and surprise-driven tone, reflecting the constraints of the domestic media environment.

Keywords: multilingual sentiment classifier, XLM-roBERTa, NLP Town, sentiment analysis, NRC emotion lexicon, LDA topic modelling, media framing, Operation Absolute Resolve, Trump–Venezuela, MediaCloud

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1 Introduction

1.1 Background

In the early hours of 3 January 2026, the United States launched a military operation in Venezuela codenamed. US Armed Forces conducted airstrikes on military infrastructure across northern Venezuela and deployed a special operations force that raided the presidential compound in Caracas, resulting in the capture of president Nicolás Maduro and his wife, Cilia Flores *Operation Absolute Resolve* (Council on Foreign Relations, 2026).

Both were transported to New York City, where they were arraigned on narcoterrorism and drug trafficking charges in Manhattan federal court on 5 January, pleading not guilty. The Trump administration justified the operation as a law-enforcement action undertaken under the president's "inherent constitutional authority", while United Nations officials, international law experts, and numerous governments condemned it as a violation of Venezuelan sovereignty and the UN Charter (Foundation for Defense of Democracies, 2026).

The operation was the culmination of a months-long escalation that had begun in September 2025, when the United States launched Operation Southern Spear — a naval and aerial military buildup in the Caribbean Sea nominally targeting drug trafficking networks (Foundation for Defense of Democracies, 2025). By December 2025, US forces had begun intercepting and seizing Venezuelan oil tankers and conducting the first direct strikes on Venezuelan soil. In parallel, Venezuelan opposition leader María Corina Machado, who had received the Nobel Peace Prize in 2025 (Norwegian Nobel Committee, 2025), publicly supported US military action to remove Maduro (Wikipedia contributors, 2026), a position that deepened regional divisions over the legitimacy of the intervention.

In the immediate aftermath of the 3 January strikes, President Trump declared that the United States would "run" Venezuela until a political transition could be arranged (Reuters, 2026). On 8 January 2026, the Venezuelan government announced the release of political prisoners as a "gesture" of cooperation; Trump subsequently announced on 9 January that a planned second wave of attacks had been canceled in response to said collaboration (The Guardian, 2026). These rapidly evolving developments, spanning the period of 2 to 10 January 2026, constitute the event window examined in this study.

International reactions were sharply polarised. Brazil, Colombia, Mexico, Chile, Spain, and Uruguay issued a joint statement condemning the military action. Colombia's President Gustavo Petro mobilised security forces along the border and threatened armed resistance. Cuba declared two days of national mourning. Meanwhile, most NATO member states and Israel expressed support or muted approval for the operation. The intervention thus

produced one of the most significant fractures in political opinion in recent decades — and generated an immediate and intense wave of international media coverage (Wikipedia, 2026).

1.2 Motivation

While the factual occurrence of the intervention is undisputed, its interpretation and description have been highly contested in public and media discourse. News outlets, political actors, and online publics have alternately described the event as an invasion, military operation, law enforcement action, or kidnapping, each term embedding distinct legal, moral, and political implications.

The use of contested descriptors allows for the systematic study of media framing, sentiment, and agenda-setting dynamics across ideologically diverse outlets and social media platforms. Understanding how language choices shape public perception is particularly relevant in highly polarized political environments.

How this crisis was *framed*, what aspects were emphasised, what emotions were evoked, and which narratives dominated, varied substantially across world regions. According to Entman (1993), media framing is never neutral: it reflects the geopolitical position, media system characteristics, and political culture of each national context. This paper asks: *How do news media across the United States, Europe, Latin America, and Venezuela differ in their framing of Operation Absolute Resolve?* To answer this question, a multi-method computational pipeline is applied to 1,000 news articles collected from MediaCloud's regional source collections, covering the period 2–10 January 2026. By combining multilingual sentiment analysis, NRC emotion scoring, bigram framing analysis, and LDA topic modelling, a comprehensive comparative portrait is produced of regional media narratives at the moment of a historically unprecedented US military intervention in Latin America.

The paper is structured as follows: Section 2 reviews the theoretical frameworks guiding the analysis. Section 3 describes the data and methods. Section 4 presents the results and discusses the findings in light of the theoretical framework. Section 5 presents limitations and Section 6 concludes with directions for future research. The full project (codes and outputs) is available at Github: CSS Project.

2 Theoretical Frameworks

2.1 Framing Theory

Framing theory (Entman, 1993) posits that media do not merely report events but actively construct meaning by selecting, emphasising, and contextualising certain aspects of reality over others. Frames organize experience and guide interpretation, shaping how audiences understand political events (Scheufele, 1999). In international political coverage, framing choices are particularly consequential: the same event can be simultaneously framed as a humanitarian crisis, a national security threat, an economic dispute, or a democratic struggle, depending on the outlet's national context and ideological orientation.

In the context of this study, framing is operationalized through three computational proxies: (1) sentiment polarity, which captures the overall evaluative tone; (2) emotional content, which reveals the affective register of coverage; and (3) recurring bigrams and LDA topics, which identify the dominant narrative themes and lexical frames employed by each regional media system.

2.2 Agenda Setting Theory

Agenda setting theory (McCombs & Shaw, 1972) argues that the media do not tell audiences *what to think*, but *what to think about*. The salience of issues in news coverage shapes public perception of their importance. At the international level, different media systems set different agendas: what is front-page news in Brasilia may be a minor story in Munich, and vice versa. The topics and themes identified through LDA topic modelling in this study provide a computational approximation of each region's news agenda with respect to the Trump–Venezuela crisis, allowing us to compare which aspects of the conflict each region prioritises.

3 Data and Methods

3.1 Data Sources and Collection

News articles were collected via the MediaCloud API, an open-source platform indexing millions of online news stories from thousands of sources worldwide (Roberts et al., 2021). The search query `article_title:(Venezuela OR Maduro) AND Trump` was applied to four regional source collections, covering the period 2–10 January 2026. Figure 1 below shows the amount of articles matching the keywords over the selected days, while Figure 2 shows the total query matches for the whole period.

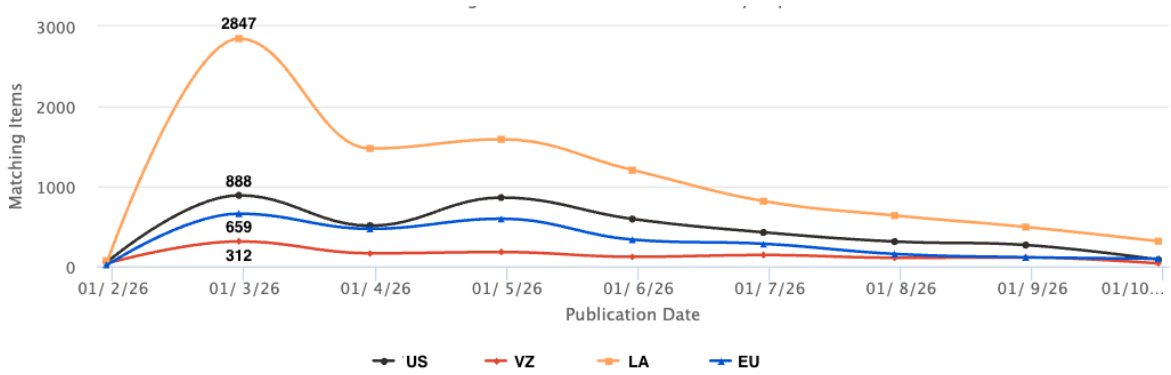

Figure 1: Total attention over time. Source: MediaCloud

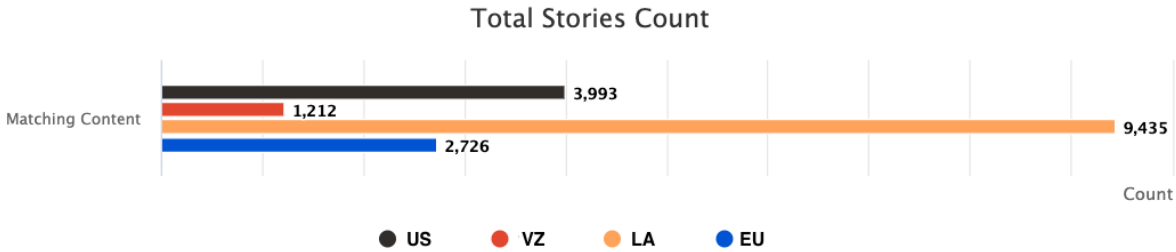

Figure 2: Total of articles per region. Source: MediaCloud

From a total of 17.366 research results, 1.000 articles were collected, 250 per region, as detailed in Table 1.

Table 1: Data collection summary by region

Region	**Collections**	**Language(s)**	**Available**	**Collected**
United States	US	English, Spanish	3.993	250
Europe	DE, FR, UK	German, French, English	2.726	250
Latin America	Brazil, Mexico	Portuguese, Spanish	9.435	250
Venezuela	Venezuela	Spanish	1.212	250
Total			17.366	**1.000**

For the sake of simplicity and directness, title-only search strategy (adding the command `article_title:` as prefix) was adopted to ensure that only articles where the central topic of Trump and Maduro as main characters were included, minimizing the retrieval of tangentially related content.

3.2 Methods

- Text preprocessing: deduplication, tokenization, lemmatization, and filtering.
- Emotional analysis: model-based sentiment scoring and lexicon-based emotional tone classification.
- Framing and Tematic analysis: bigram and latent dirichlet allocation

3.2.1 Language Detection and Translation

Article languages were detected using the `langdetect` library. Given that the NRC Emotion Lexicon is English-only, non-English titles (German, Spanish, Portuguese, French) were translated to English prior to NRC scoring, Bigram and LDA Analysis using the `deep-translator` library's Google Translate interface. Multilingual sentiment analysis (Section 3.2.2) was performed on the original untranslated titles to avoid translation artifacts in the sentiment scoring stage.

3.2.2 Sentiment Analysis

Sentiment polarity was computed using two models:

- `nlptown/bert-base-multilingual-uncased-sentiment` (NLP Town, 2026) This model is based on multilingual BERT (Warner et al., 2024)a bert-base-multilingual-uncased model for sentiment analysis fine tuned on product

reviews in six languages: English, Dutch, German, French, Spanish, and Italian. It predicts the sentiment of the review as a number of stars (between 1 and 5). This model is intended for direct use as a sentiment analysis model for product reviews in any of the languages above using a five-star rating scheme.

- `cardiffnlp/twitter-xlm-roberta-base-sentiment`
  Built on XLM-roBERTa (Barbieri, Espinosa Anke, & Camacho-Collados, 2022; Cardiff NLP, 2026b), this model is a cross-lingual transformer architecture fine-tuned on 198 million tweets across more than 30 languages as part of the TweetEval benchmark (Barbieri, Camacho-Collados, et al., 2022; Cardiff NLP, 2026a). The model classifies each headline as *positive*, *negative*, or *neutral*.

The models were selected over English-only alternatives (e.g., VADER) due to the multilingual nature of the sample. Both models are therefore capable of processing text in English, Spanish, Portuguese, French and German without requiring prior translation, making them suitable candidates for this study.

3.2.3 Emotion Analysis

Emotional tone was measured using the NRC Word Emotion Association Lexicon (Mohammad & Turney, 2013), which maps words to eight discrete emotions: anger, fear, joy, trust, sadness, surprise, disgust, and anticipation. Scores represent the average number of emotion-bearing words per headline within each region. For this purpose, non-English headlines were translated to English prior to scoring.

3.3 Framing Analysis: Bigrams

Bigram analysis was performed on translated headlines using NLTK (Bird et al., 2009), with English, Spanish, French, German, and Portuguese stopwords removed. The top 10 most frequent two-word phrases per region were extracted to identify dominant lexical frames.

3.3.1 Topic Modelling: LDA

Latent Dirichlet Allocation (LDA) identifies latent thematic clusters within the sample, providing a data-driven characterization of each region's news agenda. LDA (Blei et al., 2003) was applied to translated headlines per region using `scikit-learn` (Pedregosa et al., 2011), with $k = 5$ topics per region.

4 Results

4.1 Sentiment Polarity

The following sentiment analysis shows the sentiment distribution across regions for both models: NLP Town BERT multilingual (Table 2) and XLM-roBERTa (Table 3).

Table 2: NLP Town sentiment distribution by region (% of articles)

Region	**Negative (%)**	**Neutral (%)**	**Positive (%)**
Europe	64.8	5.6	29.6
Latin America	66.8	4.8	28.4
United States	60.8	4.0	35.2
Venezuela	64.0	3.2	32.8

NLP Town almost eliminates neutral (Figure 3), forcing it nearly every headline into positive or negative, which inflates both poles artificially, while XLM-roBERTa produces a rich neutral category (33-61%) showing it's genuinely uncertain when headlines are factual.

Table 3: XLM-roBERTa Sentiment distribution by region (% of articles)

Region	**Negative (%)**	**Neutral (%)**	**Positive (%)**
Europe	54.8	39.6	5.6
Latin America	52.4	38.8	8.8
United States	61.2	34.0	4.8
Venezuela	32.8	61.2	6.0

In XLM-roBERTa is noticeable that negative sentiment dominates in all four regions, but the degree and pattern vary substantially. The United States exhibits the most pronounced negative coverage (61.2%), combined with the lowest neutral rate (34.0%), indicating a highly evaluative and polarised domestic media environment. Europe and Latin America show near-identical distributions, both strongly negative, suggesting a convergent critical framing of the conflict among geographically and politically distinct but similarly positioned observers. Venezuela is the notable outlier: it records the highest neutral rate (61.2%) and the lowest negative rate (32.8%), consistent with a more restrained, factual reporting style typical of constrained media environments (see Figure 4).

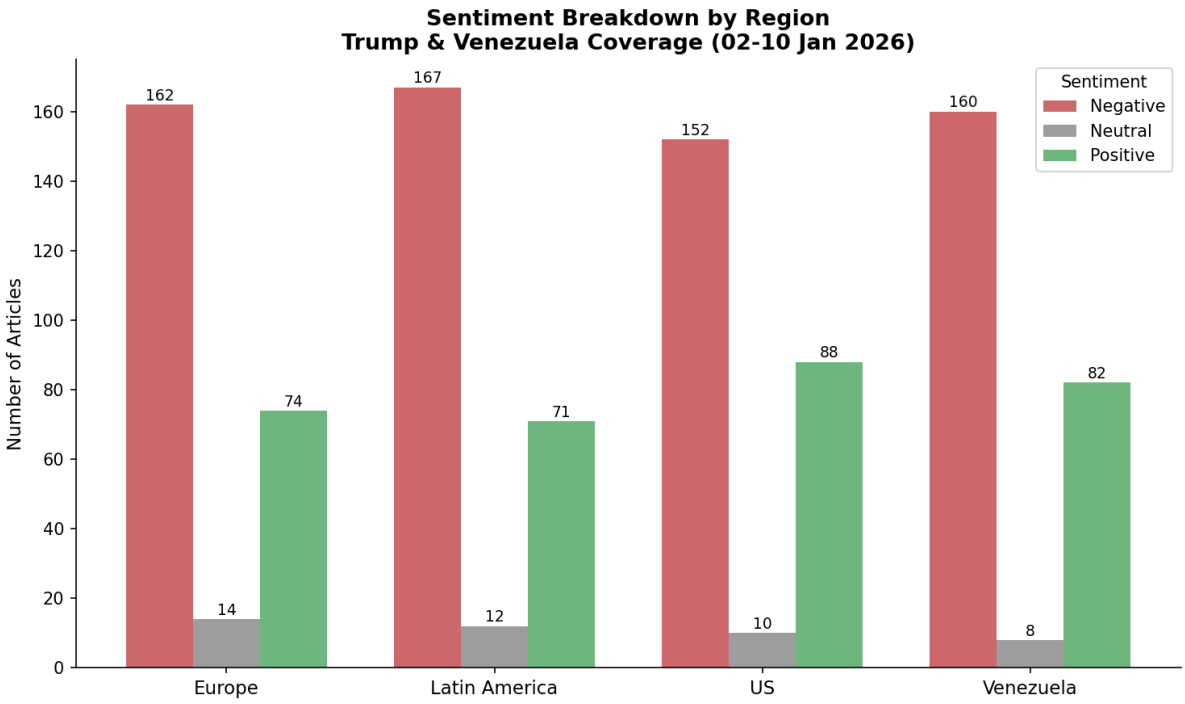

Figure 3: NLP Town BERT sentiment analysis

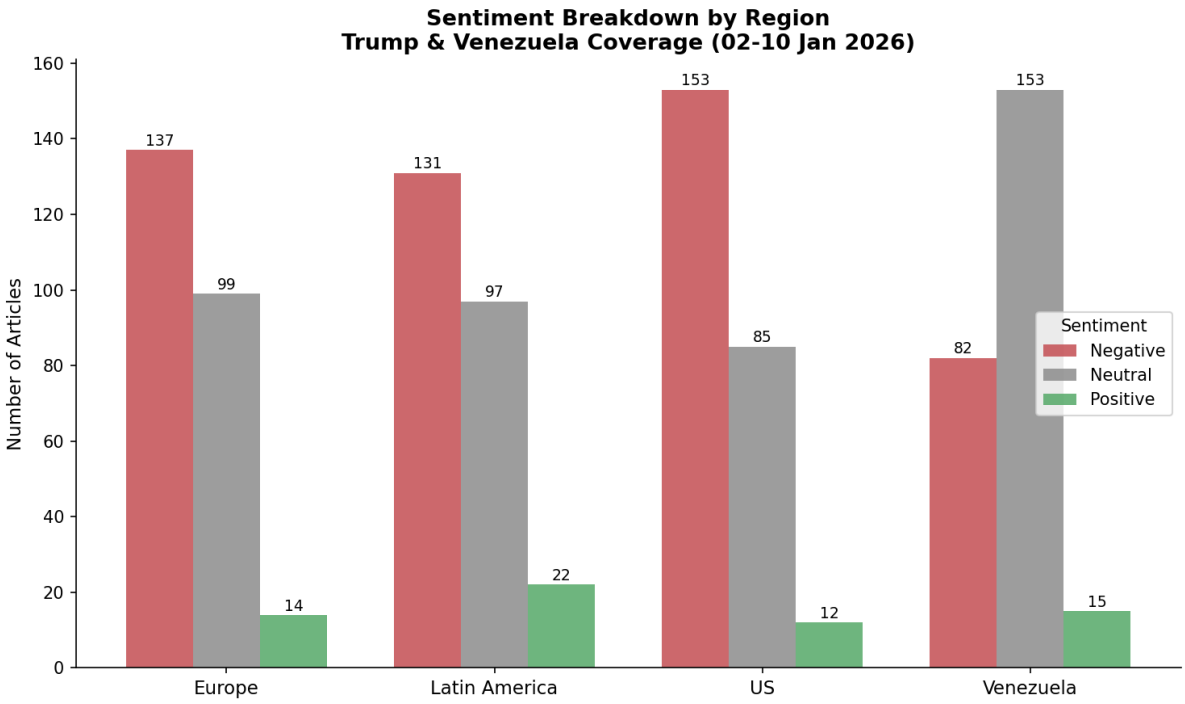

Figure 4: XLM-roBERTa sentiment analysis

Model Selection: Robustness Check

Both models tested in this study were selected on the basis of their native multilingual capabilities, which was a primary requirement given the four-language composition of the sample. The key difference lies in their training domain and output format: XLM-roBERTa produces three-class sentiment labels (positive, negative, neutral) derived from social media data (Cardiff NLP, 2026b), while NLP Town maps text to a star-rating scale originally designed for consumer reviews (NLP Town, 2026). To assess the sensitivity of sentiment results, both models were applied to the full dataset and their outputs compared. Results diverged substantially: NLP Town reduced the neutral category to near-zero across all regions (2–6%), redistributing interpretation primarily to the negative pole. Most consequentially, it reversed the most theoretically significant finding of the study: Venezuela's predominantly neutral coverage (61.2% under XLM-roBERTa) collapsed to 3.2% neutral under NLP Town, effectively rendering Venezuelan media indistinguishable from all other regions in terms of sentiment distribution. NLP Town also produced higher positive sentiment scores across all regions, an artefact likely attributable to its star-rating training objective, in which mid-range ratings (3–4 stars) skew positive relative to the neutral category in news text. The discrepancies can be attributed to this domain mismatch. NLP Town was trained on consumer review platforms like Amazon, Yelp, and TripAdvisor, where language is evaluative by design and a neutral tone is comparatively rare. News headlines, by contrast, regularly employ factual, descriptive language that carries no clear evaluative valence and should correctly be classified as neutral. XLM-roBERTa, trained on short opinionated social media texts, represents a closer domain match to news headlines and preserves the theoretically meaningful neutral signal (Barbieri, Camacho-Collados, et al., 2022).

4.2 Emotional Tone

Mean NRC emotion scores by region are presented on Table 4. Several patterns can be noted: **Fear** is the dominant emotion across all regions, but is particularly pronounced in the United States (0.616) and Latin America (0.524), suggesting threat-based framing in both the country initiating pressure and its immediate regional neighbors. **Anger** is highest in the United States (0.456), consistent with a politically charged domestic discourse observed on the headlines. **Trust** is notably high in US coverage (0.532), which seems counterintuitive at the beginning, but on the other hand may reflect frames of institutional legitimacy surrounding Trump's executive actions. **Surprise** reaches its peak in Venezuelan coverage (0.908), far exceeding all other regions, indicating a reactive, event-driven reporting style in which headlines consistently signal the unexpected nature of

developments. Europe presents the most balanced emotional profile, with moderate scores across all dimensions, consistent with a detached, analytical framing style (see Figures 5 and 6).

Table 4: Mean NRC emotion scores by region

Region	**Anger**	**Fear**	**Joy**	**Trust**	**Sadness**	**Surprise**	**Disgust**	**Anticip.**
US	0.456	0.616	0.080	0.532	0.168	0.508	0.164	0.204
Europe	0.304	0.420	0.140	0.428	0.196	0.656	0.116	0.272
Latin America	0.288	0.524	0.172	0.416	0.240	0.720	0.092	0.256
Venezuela	0.196	0.308	0.172	0.532	0.100	0.908	0.092	0.276

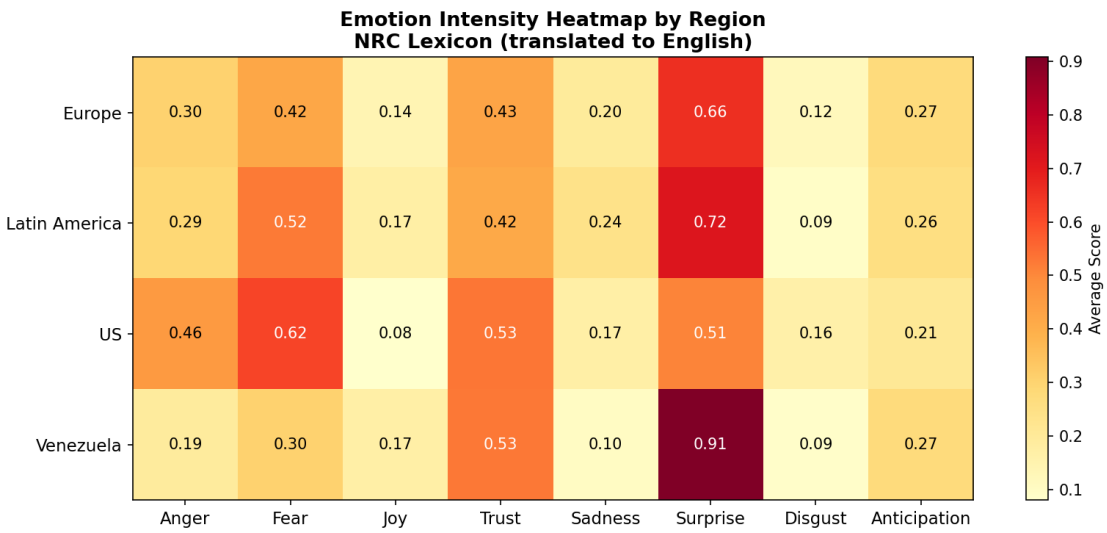

Figure 5: Emotions heatmap per geographical region.

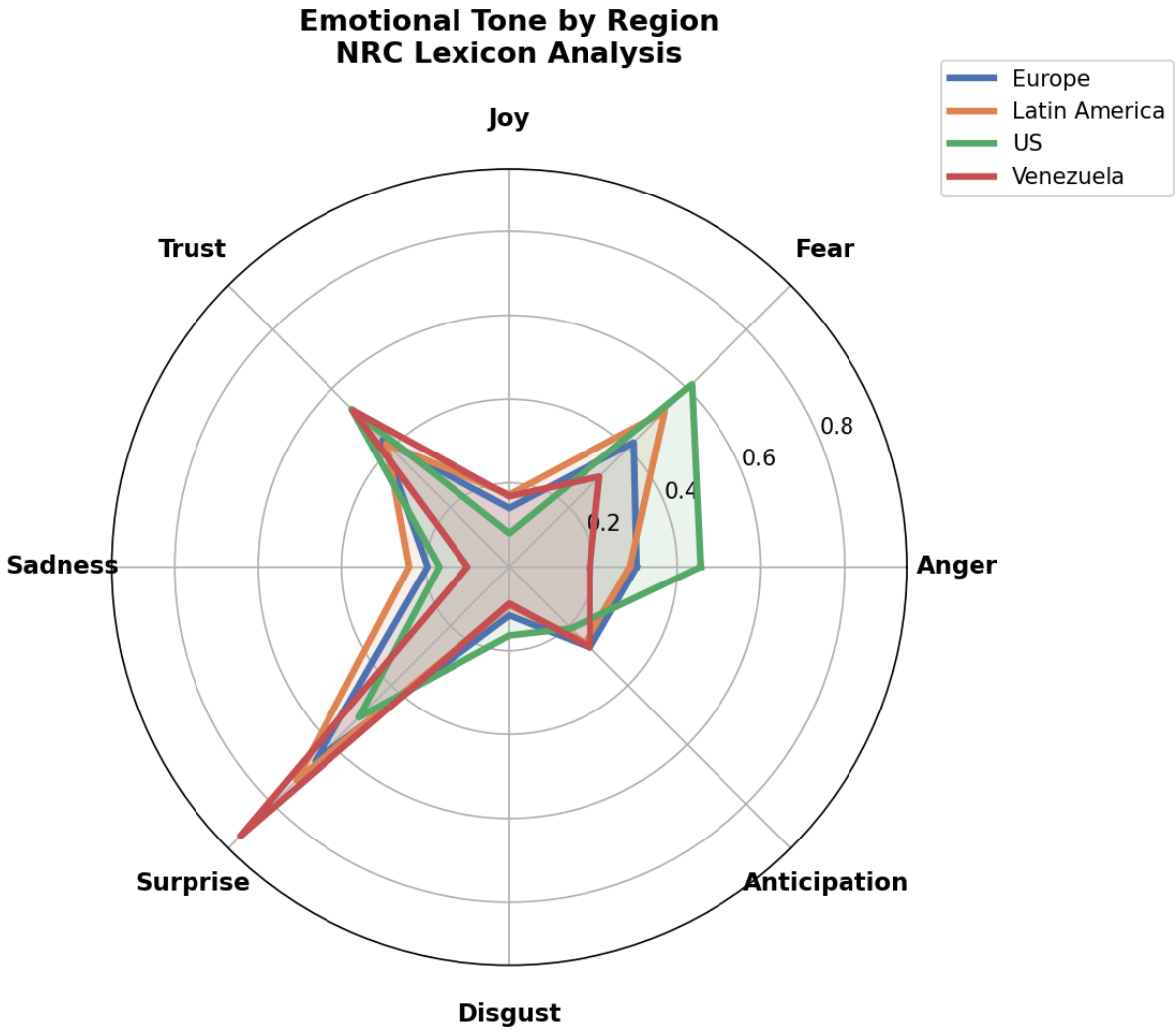

Figure 6: Emotions radar chart per geographical region.

4.3 Bigram Framing Analysis

Bigram analysis reveals distinct lexical frames across regions (see Figure 7). **European** coverage centres on "donald trump" (37) and economic framing ("oil companies", "invest venezuela"), while uniquely connecting Venezuela to Trump's other foreign policy moves ("venezuela greenland", "greenland ukraine") — situating the crisis within a broader pattern of US expansionism and claims from Trump concerning possible next targets for annexation. **Latin America** foregrounds economic sovereignty ("venezuelan oil" 20, "protect venezuelan" 12, "oil revenues" 12) and humanitarian concerns ("political prisoners" 9), reflecting a regional perspective sensitive to intervention and resource control. **US** headlines are dominated by action-oriented regime change language: "capture maduro" (21), "run venezuela" (25), "strikes venezuela" (11), framing the conflict as a direct intervention

operation. **Venezuelan** coverage combines domestic political actors ("corina machado", "donald trump") with economic concerns ("oil companies", "venezuelan oil"), suggesting a dual focus on internal political dynamics and external economic threats.

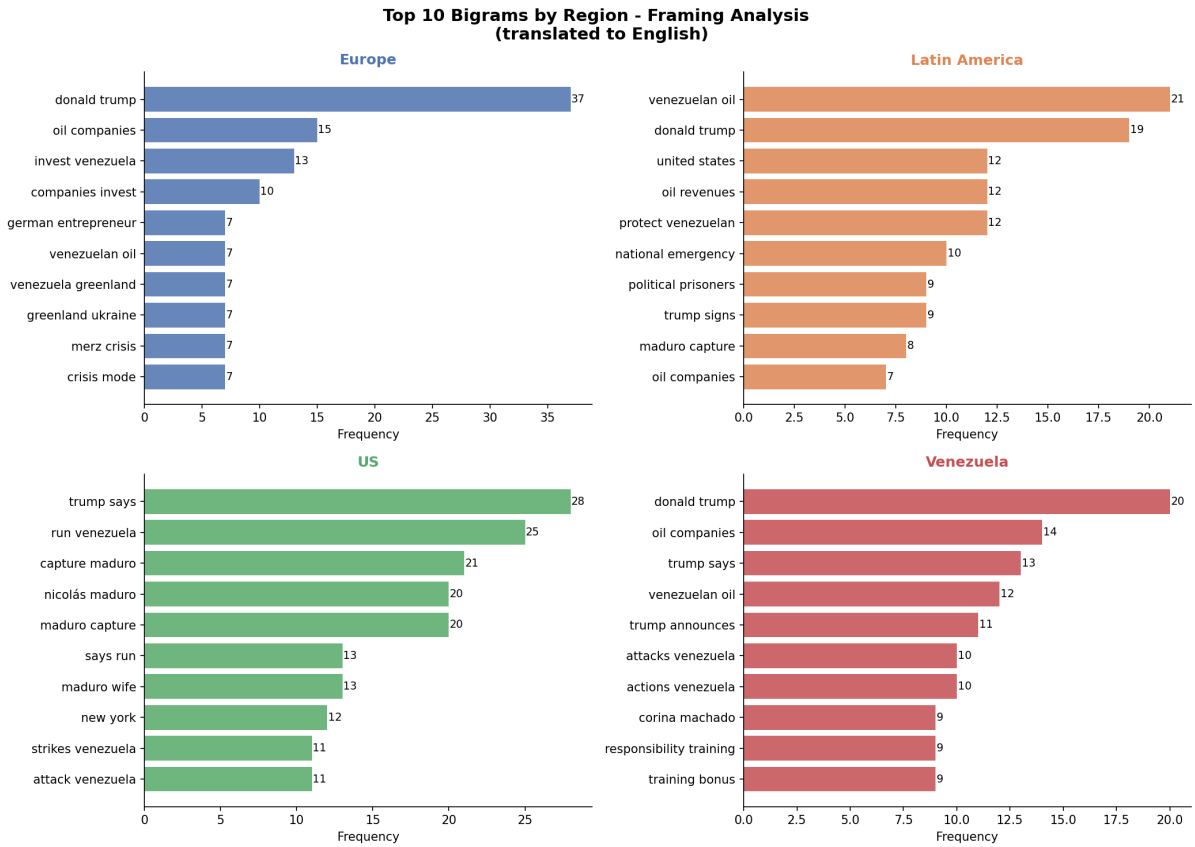

Figure 7: Bigrams per region.

4.4 LDA Topic Modelling

LDA topic modelling reveals three coherent topic clusters per region (see Figure 8). In **Europe**, topics reveal a geopolitical contextualization strategy, linking Venezuela to Greenland, Ukraine, and German domestic politics, framing the crisis as part of Trump's broader international posture. **Latin America** produces topics naming government leaders and regime change narrative, reflecting internal reorganization, suggesting a regional diplomatic response framing. In the **United States**, all three topics converge the capture of Maduro, military pressure and centering in oil investments. **Venezuela's** topics consistently foreground oil protection, executive orders, indicating a state-aligned narrative focused on economic resilience and geopolitical alignment.

LDA Topic Modeling by Region (translated to English)

Europe

Topic	Top Words
Topic 1	venezuela, oil, trump, companies, invest
Topic 2	trump, donald, america, order, venezuelan
Topic 3	trump, greenland, venezuela, ukraine, germany
Topic 4	maduro, trump, capture, says, ready
Topic 5	trump, venezuela, greenland, german, maduro

Latin America

Topic	Top Words
Topic 1	trump, nobel, prize, machado, corina
Topic 2	trump, oil, venezuelan, venezuela, protect
Topic 3	trump, venezuela, government, political, new
Topic 4	trump, meeting, 2026, saturday, january
Topic 5	maduro, venezuela, trump, capture, oil

US

Topic	Top Words
Topic 1	maduro, venezuela, federal, charges, trump
Topic 2	trump, venezuela, says, run, maduro
Topic 3	maduro, venezuela, trump, captured, venezuelan
Topic 4	maduro, capture, venezuela, trump, military
Topic 5	venezuela, trump, operation, calls, react

Venezuela

Topic	Top Words
Topic 1	trump, venezuelan, oil, bonus, protect
Topic 2	trump, venezuela, says, machado, donald
Topic 3	trump, petro, military, white, house
Topic 4	trump, venezuela, senate, military, new
Topic 5	trump, venezuela, oil, companies, second

Figure 8: Top 3 *hot-topics* per region.

4.5 Discussion

The results provide strong empirical support for the core predictions of framing theory and agenda setting theory in the context of international political communication.

Consistent with framing theory by Entman (1993), *geopolitical proximity shapes emotional intensity*, media systems frame events through the lens of their own national interests and geopolitical positions. US media, closest to the political source of the crisis, exhibit the most emotionally intense coverage (high fear, anger, and anticipation), framing the conflict as a high-stakes political operation. Latin American media, geographically and politically proximate to Venezuela, show the highest fear scores after the US, reflecting anxieties about regional stability and the precedent of US intervention. European media, more distant from the immediate conflict, adopt a more analytical and factual frame.

Venezuelan media constraint is computationally detectable. The exceptionally high neutral sentiment rate (61.2%) and dominant surprise emotion (0.908) in Venezuelan coverage are consistent with theoretical accounts of state-influenced media systems, in which outlets report events descriptively rather than evaluatively (Hallin & Mancini, 2004). The computational signature of constrained media — factual, reactive, emotionally flat except for surprise — is clearly visible in the data, offering a methodological contribution to the computational study of media systems.

Following McCombs and Shaw (1972), *agenda setting produces distinct topical emphases.* This means, different media systems prioritise different aspects of the same event. European media uniquely set an agenda connecting Venezuela to Greenland and Ukraine, situating the crisis within a frame of Trump's global territorial ambitions. Latin American media set an agenda of oil sovereignty and diplomatic response. US media set a regime-change agenda. These divergent agendas likely shape public understanding of the conflict differently across regions.

A methodological contribution of this study is the application of machine-translation NRC emotion scoring to a multilingual dataset. Prior to translation, Spanish and Portuguese headlines scored near-zero on all emotion dimensions, masking genuine emotional content. Post-translation, Latin American and Venezuelan emotion profiles became substantively interpretable, with Latin America revealing the highest fear score in the dataset. This approach has proven to be a pragmatic alternative to multilingual emotion lexicons.

On Figure 9 the benefit of machine-translation is visible, by direct comparison of terms across all four regions. The impact is immediately noted on the word **attack**, which is more emphasized in Venezuelan and Latin American contexts, still present in European region and considerably smaller in the US. The words **capture** and **strike** gain more prominence, suggesting a framing of justified law-enforcement actions, corroborating the previous results.

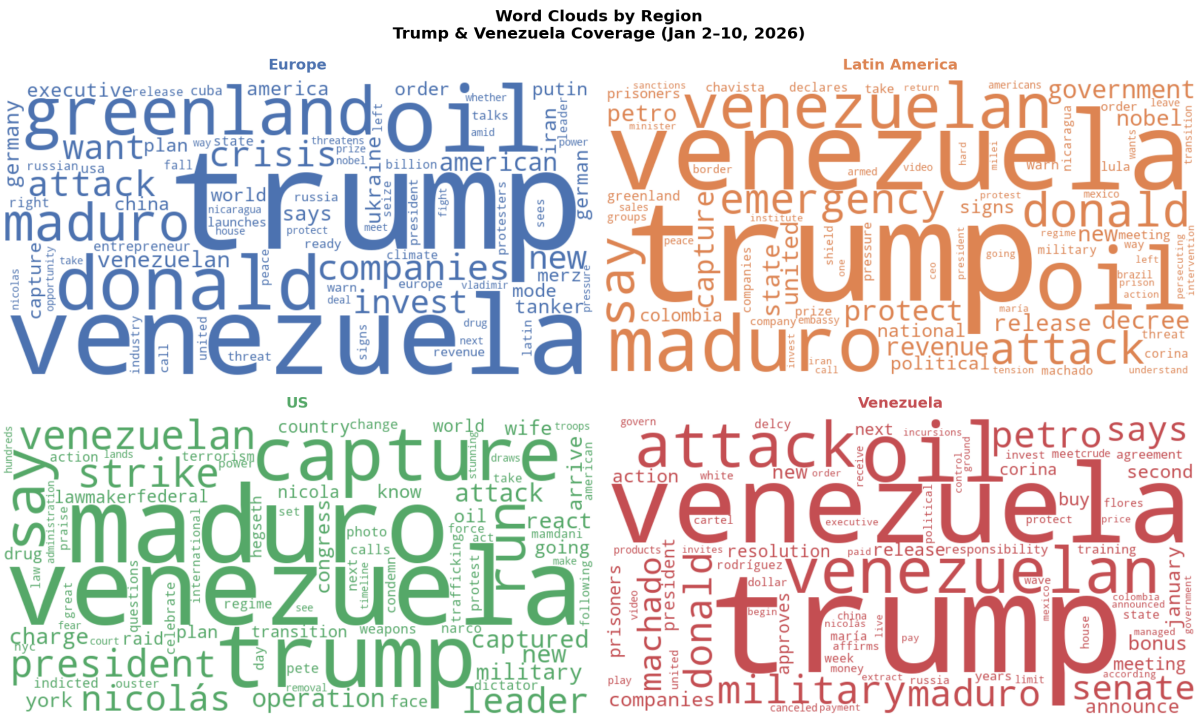

Figure 9: Wordclouds per region.

5 Limitations

This study recognises several limitations related to data collection, technical infrastructure, analytical methods, and ethical considerations.

Query design and sampling bias. The search query was restricted to article headlines containing the terms Trump, Maduro, or Venezuela. While this approach ensured topical relevance, it also means that these same words appear with high frequency across all analytical methods (including bigrams, LDA topic modelling, and wordclouds) not because they are organically salient, but because they were built into the search criteria. Results from frequency-based analyses should therefore be read as describing how these actors are contextualised in coverage, rather than as neutral measures of word importance. A further source of noise is the common practice of appending keywords to headlines for search engine optimisation, which may have caused some articles to match the query without being substantively relevant to the topic.

MediaCloud API constraints. Collecting data through MediaCloud's free tier presented several practical challenges. The platform limits users to 1,000 API calls and 20,000 stories per week, a threshold that was reached multiple times during the iterative process of query testing and debugging. The API also experienced frequent connection timeouts and empty responses, which required the implementation of automatic retry logic and waiting periods between requests. Because of these constraints, it was not possible to collect data from all desired countries in a single run, and geographic coverage was limited to a few source collections per region, selected based on the depth of their representation in MediaCloud. Future research would benefit from institutional API access.

Encoding and language specific characters. The decision to cap the dataset at 250 articles per region, 1.000 articles in total, was shaped by two factors. First, the API limitations described above made larger-scale collection impractical within the project timeframe. Second, handling text in four different languages simultaneously created encoding difficulties involving character sets, accents, and web-specific symbols that could not be fully resolved through automated cleaning alone, requiring partial manual intervention. This placed an upper bound on the volume of data that could be reliably processed. It is also worth noting that each region is represented by a small number of national outlets, which limits how broadly the findings can be generalized across the full media landscape of each region.

Sentiment model accuracy and multilingual framing. The performance of sentiment classification models varies depending on the language and type of text they were trained on. As shown in the model comparison presented in Results (Section 4), different models can produce markedly different results for the same dataset, with real

consequences for how findings are interpreted. The NRC Emotion Lexicon presented an additional challenge, as it is designed for English text only. Non-English headlines were therefore machine-translated before emotion scoring, a process that may introduce inaccuracies. Beyond technical accuracy, politically charged terms such as "capture", "intervention", or "attack" carry different connotations across languages and cultural contexts — a layer of meaning that automated tools are not always able to capture reliably.

Ethical considerations. This study examines coverage of a politically sensitive and still-evolving geopolitical event. All findings are based on aggregate patterns across the dataset, and no claims are made about the intentions or editorial choices of individual news outlets. Throughout the analysis and interpretation, care was taken to maintain a balanced and descriptive approach, avoiding the reinforcement of extreme positions or the uncritical reproduction of politically charged language.

6 Conclusion

This study demonstrates that regional media framing of the Trump–Venezuela conflict in early January 2026 reflects each region's geopolitical position, media system characteristics, and political culture. Using a multilingual computational pipeline combining XLM-roBERTa sentiment analysis, NRC emotion scoring, bigram framing analysis, and LDA topic modelling, four distinct regional media narratives can be identified: a regime-change frame (US), a geopolitical contextualisation frame (Europe), a sovereignty and humanitarian frame (Latin America), and a reactive, neutral, surprise-driven frame (Venezuela).

Several limitations should be noted. First, the sample of 250 articles per region, while sufficient for exploratory analysis, limits statistical generalisation. Future work should employ larger sample. Second, machine translation introduces potential artifacts in emotion scoring, though is arguably preferable to the alternative of ignoring non-English content entirely. Third, LDA with $k = 5$ topics is a simplification; sensitivity analyses with varying k values are recommended.

Despite these limitations, the study illustrates the value of multilingual computational methods for comparative international communication research, and contributes empirically to the study of how geopolitical crises are differentially framed across the world's media systems.

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